

ITAI 1371 – Final ML Project

Topic: Song Popularity Prediction Using Spotify Audio Features

Step 1 — Interest Area

Music Streaming Analytics and Understanding What Makes Songs Popular

Step 2 — Three Possible ML Project Ideas

Machine learning projects usually do one of five things: predict a number, classify, detect, recommend, or group similar things. Here are three ideas I came up with for this topic:

- “Can ML predict a song’s popularity score on Spotify using audio features like danceability, energy, and valence?”
- “Can ML classify songs into Hit, Moderate, or Flop categories based on their audio features?”
- “Can I detect hidden groups of similar songs by clustering Spotify’s audio features?”

Step 3 — State-of-the-Art Research

Source 1 — Research Paper (Google Scholar)

Paper: *Predicting song popularity based on Spotify’s audio features: insights from the Indonesian streaming users* (Journal of Management Analytics, 2023)

1. What are they trying to solve?

They try to figure out how Spotify’s audio features influence whether a song becomes popular, and they focus on Indonesian listeners as a case study.

2. What data did they use? Can I access it?

They used a public Kaggle Spotify dataset with around 92,000 songs and 20 features. Yes, the same type of dataset is available on Kaggle and I can download it for free.

3. What ML method did they use?

They compared several regression and classification models. Extra Trees Regressor worked best for predicting the popularity score, and Random Forest Classifier was best for sorting songs into popularity tiers.

4. One important result:

Audio features alone can explain a meaningful part of why certain songs become hits, but region and listener culture also play a big role, so the model doesn’t work equally well in every market.

Source 2 — Trusted Article / Blog

Article: Spotify Engineering – *How Spotify Uses ML to Power Discover Weekly and Music Recommendations*

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1. What are they trying to solve?

Spotify wants to figure out which songs each user will like next so it can power features like Discover Weekly, Release Radar, and Daily Mix.

2. What data did they use?

They use a mix of user listening history, audio features from their internal audio analysis pipeline, playlist co-occurrence data, and interaction signals like skips and saves.

3. What ML method did they use?

A combination of collaborative filtering, content-based models on audio features, and deep learning embeddings that represent each song as a vector.

4. One important result:

Combining audio-based features with user behavior data gives much better recommendations than using either one alone, which is why their hybrid approach works so well.

Step 4 — Three Big Companies in This Area

1. Spotify

- **Product / Technology:** Audio streaming platform with ML-powered personalization, including Discover Weekly, Daily Mix, and Release Radar.
- **How ML helps:** It personalizes each user's home page, picks playlist recommendations, and decides which tracks get surfaced first.
- **Interesting detail:** Discover Weekly refreshes every Monday and is generated individually for each of their users based on similar listener patterns.

2. Apple Music

- **Product / Technology:** Music streaming service that blends human editors with ML models for curated playlists and personal stations.
- **How ML helps:** ML ranks songs for the For You section and powers the autoplay / station feature after a track finishes.
- **Interesting detail:** Apple bought Shazam back in 2018 and uses its audio fingerprinting data to improve how songs are matched and tagged.

3. TikTok (ByteDance)

- **Product / Technology:** Short-video app where the background song often decides if a clip goes viral, making it a huge force in music discovery.
- **How ML helps:** The recommendation engine on the For You page picks which sounds appear in front of users, which directly pushes certain songs up the charts.
- **Interesting detail:** A lot of songs now blow up on TikTok first before they hit the Spotify or Billboard charts, which has changed how labels release music.

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Step 5 — Shrinking the Topic to Match the Course

This is the part where I make the project doable with the tools and models we have used this semester.

Dataset:

I will use the “Spotify Tracks Dataset” on Kaggle (by maharshipandya). It has about 114,000 songs across 125 genres, with Spotify audio features and a popularity score for each track. Link: kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset

Input (Features):

danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration_ms, key, mode, time_signature, explicit flag, and track genre.

Output (Target):

- Main target: popularity score (0–100) for regression.
- Secondary target: popularity tier (Low, Medium, High) for classification, made by binning the popularity score.

Algorithm Type:

Supervised learning for the main task both regression (predicting the score) and classification (predicting the tier). I also plan to use unsupervised learning (K-Means clustering) to explore groups of similar songs, which lines up with what we covered in Module 10.

Models I Plan to Try:

- Linear Regression and Logistic Regression as baselines (Module 6).
- Random Forest (Regressor and Classifier) ensemble method from Module 9.
- Gradient Boosting (XGBoost or sklearn’s GradientBoosting) for comparison, also from Module 9.
- K-Means clustering to find natural groupings of songs by their audio profile (Module 10).
- Hyperparameter tuning using GridSearchCV on the best two models (Module 11).
- Evaluation with train/test split and cross-validation: RMSE and R^2 for regression, and accuracy / precision / recall / F1 for classification (Module 7).
- I also want to briefly look at fairness whether certain genres get systematically lower predicted popularity since we talked about this in Module 12.

Step 6 — Abstract

This project looks at whether machine learning can predict the popularity of a song on Spotify using only its audio features and basic metadata. It matters because streaming platforms like Spotify, Apple Music, and TikTok rely heavily on algorithms to decide which songs get surfaced to users, so understanding the link between audio features and popularity is useful for artists, record labels, and the platforms themselves. A 2023 study on Spotify’s Indonesian market found that ensemble models like Extra Trees and Random Forest can predict popularity fairly

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well using features like danceability, energy, and valence, although cultural context also plays a role. In industry, Spotify builds hybrid models that combine audio features with user listening behavior; Apple Music mixes editorial curation with ML ranking; and TikTok uses its recommendation engine to push sounds into the For You feed, which directly affects what becomes a hit. For this course project, I will use the public Spotify Tracks Dataset from Kaggle (around 114,000 songs) and build two supervised models a regression model to predict the raw popularity score, and a classification model to sort songs into Low, Medium, and High popularity tiers. I will start with Linear and Logistic Regression baselines, then move to Random Forest and Gradient Boosting, apply K-Means clustering to explore groupings in the audio features, and tune the best models using GridSearchCV. I will evaluate using RMSE and R^2 for regression and accuracy and F1 for classification, and then compare my results with the state-of-the-art paper from Step 3.

Step 7 — Final Submission Summary

Topic: Song Popularity Prediction Using Spotify Audio Features

Includes:

- Interest area
- Three ML project ideas
- Two state-of-the-art sources (one paper, one blog)
- Three companies using ML in this area
- Dataset plan, features, target, and model choices
- One-paragraph abstract